



HACK2SKILL

BHARATIYA ANTARIKSH HACKATHON 2026



Team Name : Sentinel8

Team Leader Name : Shivansh Katiyar

Problem Statement : PS-11 Cross-Modal Satellite Image Retrieval Using Multi-Sensor Remote Sensing Data



Team Leader:

Name: Shivansh Katiyar

College: Atal Bihari Vajpayee Indian Institute of Information Technology and Management, Gwalior

Team Member-1:

Name: Prabal Poddar

College: Atal Bihari Vajpayee Indian Institute of Information Technology and Management, Gwalior

Team Member-2:

Name: Hemateja Chandaluri

College: Atal Bihari Vajpayee Indian Institute of Information Technology and Management, Gwalior

Team Member-3:

Name: Srijan Singh

College: Atal Bihari Vajpayee Indian Institute of Information Technology and Management, Gwalior



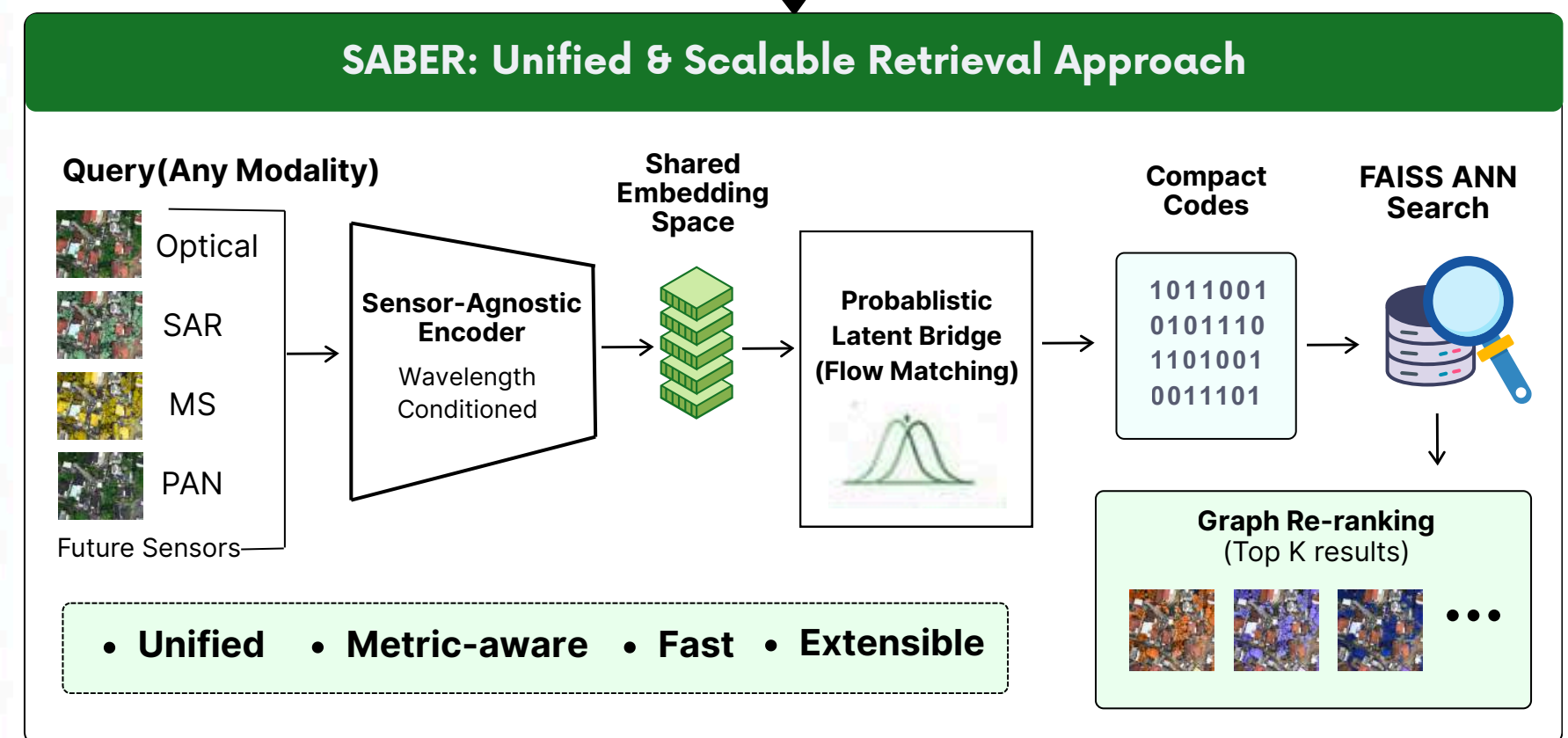
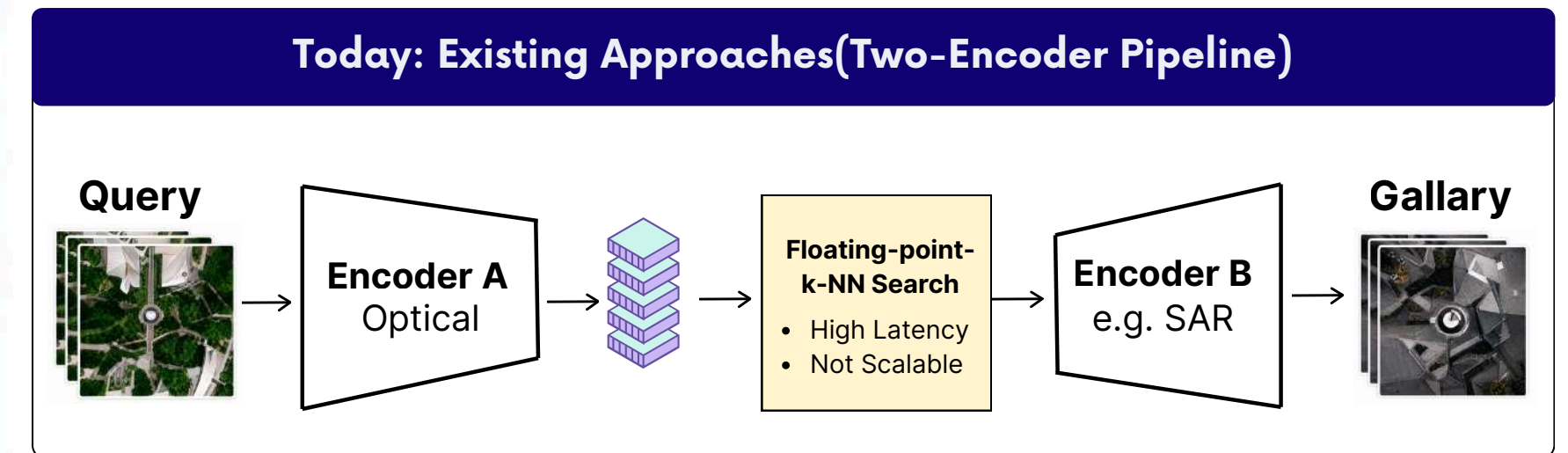
Research Document: <https://drive.google.com/file>



Github Repository: <https://github.com/SK8-infi/SABER>

	Existing Approaches	Research Gaps	SABER Innovation
Architecture	X-JEPA, REJEPA, CSMAE Separate encoder per modality	- Limited to two fixed modalities - Poor scalability to new sensors	One wavelength-conditioned encoder for all sensors (SAR, MS, PAN, HSI, future EO)
Cross-modal Alignment	Deterministic latent mapping (PSA/JEPA) One-to-one assumption	- Fails to model one-to-many correspondence - Mismatch with multi-label relevance	Probabilistic latent bridge (flow-matching) Models one-to-many cross-modal relationship
Embedding Objective	Optimized for instance similarity - Euclidean space - Metric not aligned with evaluation	- Does not reflect semantic relevance (label overlap) - Suboptimal retrieval ranking	Metric-aware embedding geometry Aligns distances with semantic relevance (multi-label overlap)
Retrieval Strategy	Brute-force floating-point k-NN search High latency at scale	- Latency not addressed - Not suitable for large archives	Compact embedding + FAISS ANN retrieval Sub-millisecond scalable search with re-ranking
Scalability & Extensibility	Trained from scratch on limited pairs (~14K) Hard to extend	- Data hungry - Not future-proof	Foundation-model backbone + LORA Extensible to more sensors with minimal adaptation

----- ARCHITECTURE COMPARISON -----



UNIQUE VALUE PROPOSITION



Sensor Agnostic & Extensible
One model for all present and future Earth observation sensors.



Metric-Aligned Retrieval
Embedding geometry aligned with semantic relevance, improving cross-modal F1.



Fast & Scalable
Compact codes + FAISS ANN enable sub-millisecond retrieval at scale.



Holistic Innovation
Novel contributions at representation, alignment and retrieval levels.

1. UNIVERSAL SENSOR SUPPORT

Today's Supported Sensors

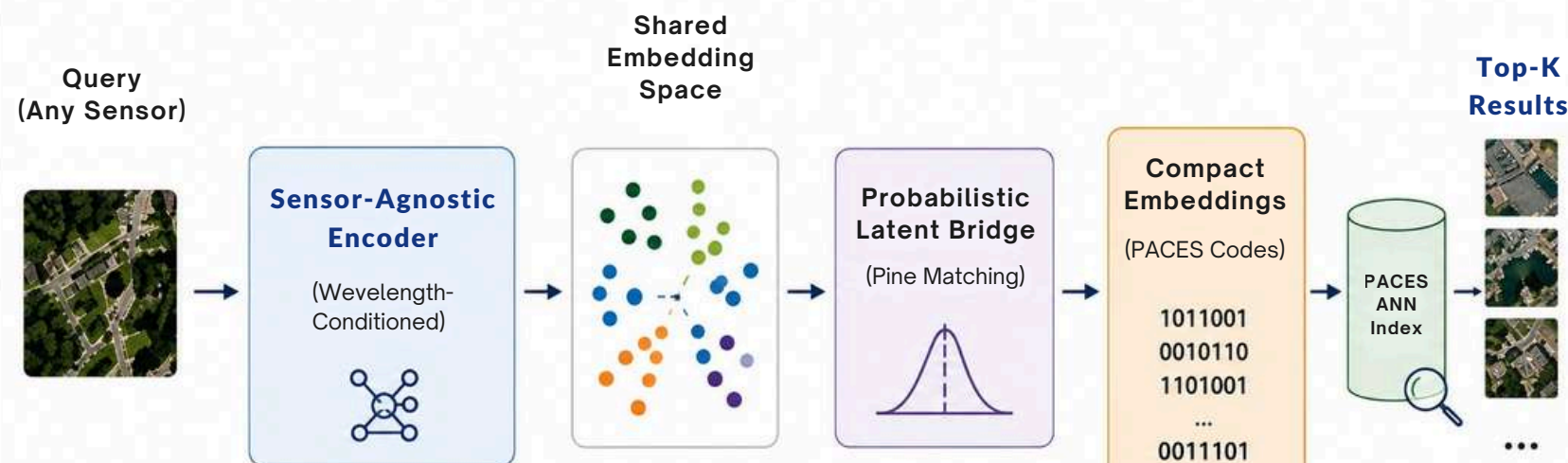


Future Sensors (Ready)



Wavelength-conditioned encoder enables seamless addition of new sensors with minimal adaptation.

2. SABER UNIFIED RETRIEVAL PIPELINE



Unified • Metric-aware • Fast • Scalable • Extensible

3. CAPABILITY COVERAGE MATRIX

Capability	X-JEPA	REJEPA	CSMAE	SABER (Ours)
Same-modal retrieval	✓	✓	✓	✓
Cross-modal retrieval	✓	✗	✓	✓
Any sensor support	✗	✗	✗	✓
Future sensor extensibility	✗	✗	✗	✓
Sub-millisecond search	✗	✗	✗	✓
Metric-aware retrieval	✗	✗	✗	✓
Foundation model backbone	✗	✗	✗	✓
Minimal retraining (LORA)	✗	✗	✗	✓
Graph re-ranking	✗	✗	✗	✓

4. RETRIEVAL MODES



Same-modal Retrieval
e.g., SAR -> SAR, MS MS



Cross-modal Retrieval
e.g., SAR -> MS, MS -> PAN



Mixed-archive Search
e.g., one query across any sensors



Top-K Ranking
Top-5/Top-10 ranked results



Semantic Retrieval
F1-tuned for label relevance

5. AI & TECHNICAL INTELLIGENCE



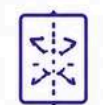
Unified Embedding Space
All sensors mapped to a common space



Probabilistic Latent Bridge
Models one-to-many cross-modal correspondence



Metric-aware Optimization
Aligns with semantic relevance (F1), not just instance similarity



Compact Representation
FAISS codes (~1M QPS) for fast, scalable retrieval



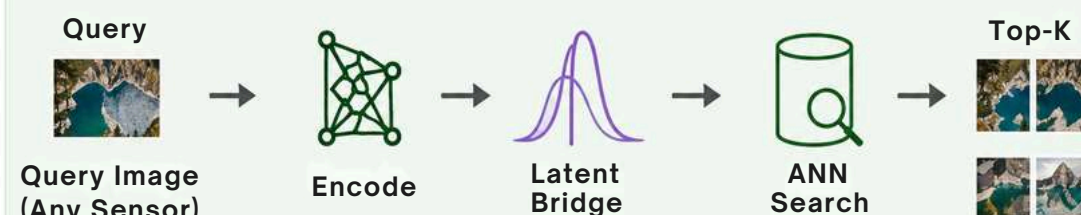
Graph Re-ranking
Refines results using multi-label relevance and topology

7. DEPLOYMENT PIPELINE

Offline (Gallery Processing)



Online (Query Time)

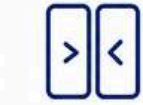


Sub-millisecond retrieval at scale

8. REAL-WORLD APPLICATIONS



Disaster Response
Rapid situational awareness



Change Detection
Multi-temporal analysis



National EO Archives
Unified access across sensors



Multi-Agency Search
Cross-organization data discovery

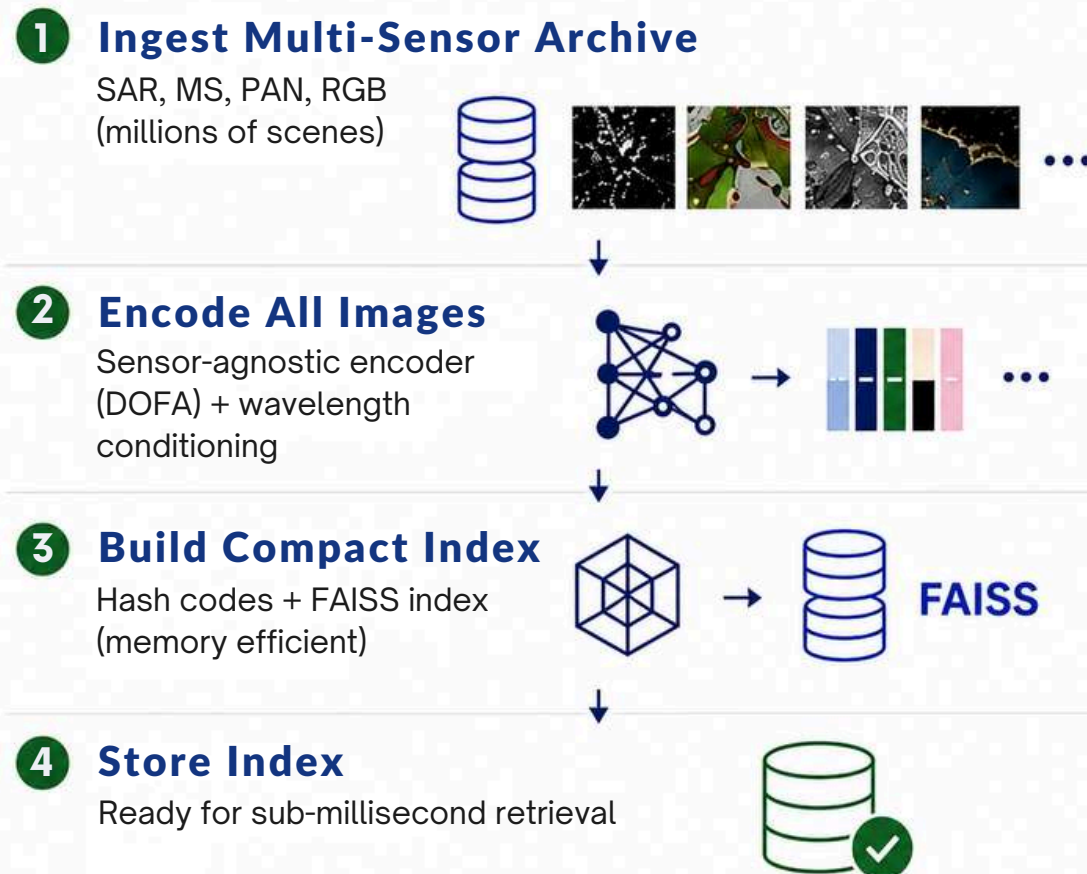


EO Analytics
Land, ocean, urban monitoring



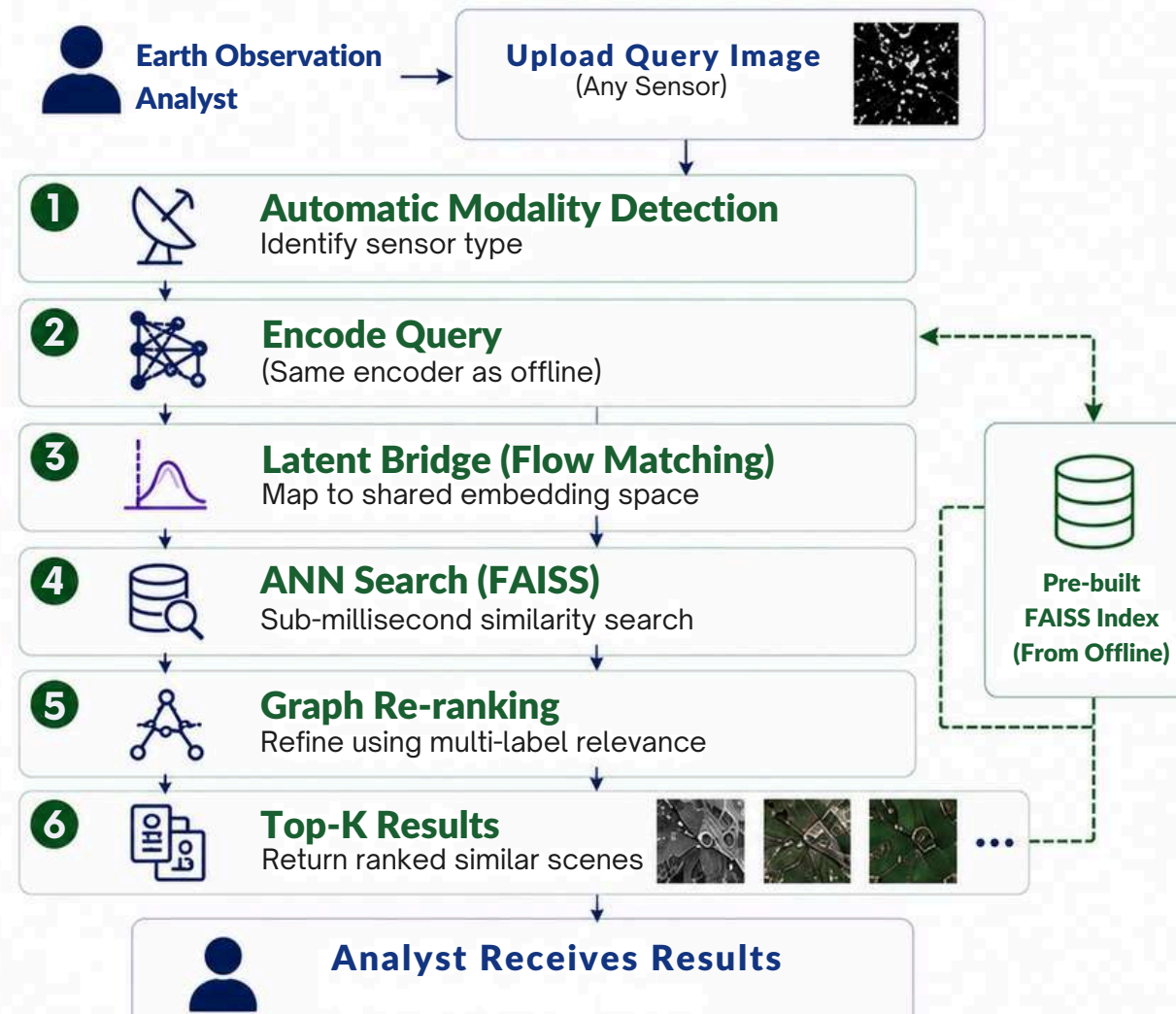
ISRO Mission Archives
Long-term mission data retrieval

1. OFFLINE PREPARATION (ONE-TIME)



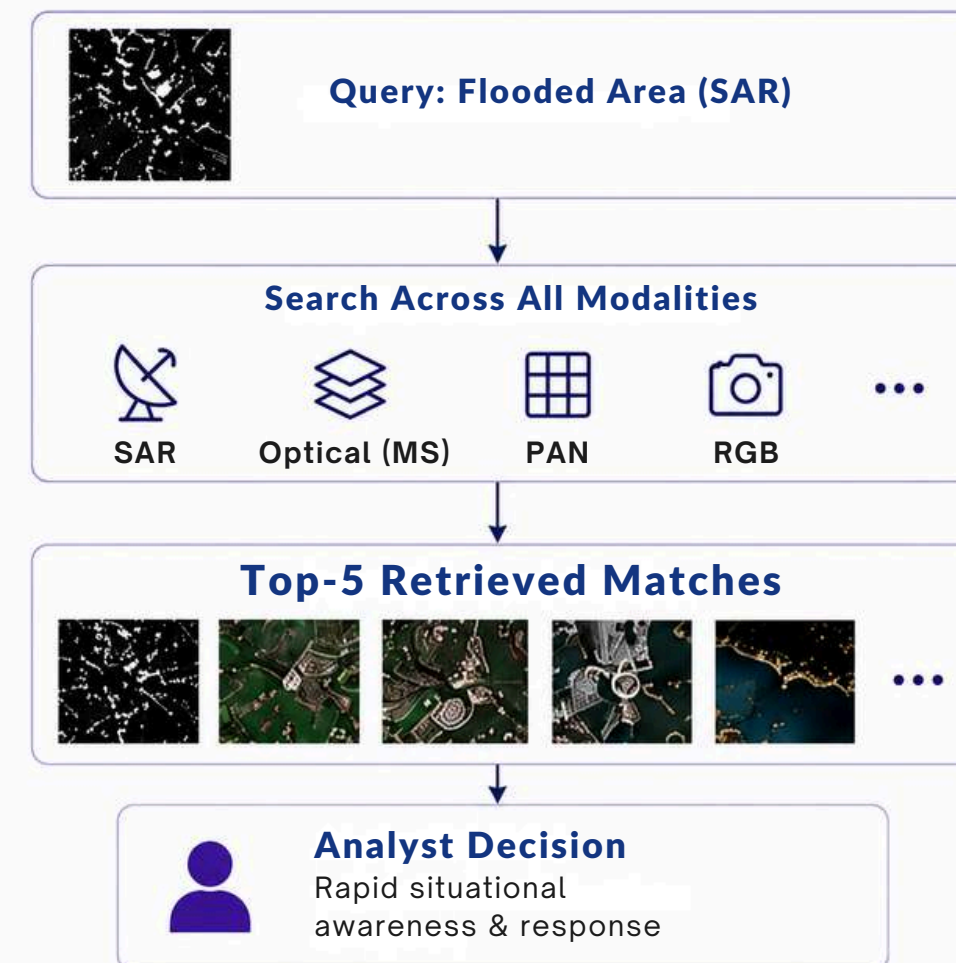
Heavy computation done once. Scales to millions of images.

2. ONLINE QUERY WORKFLOW (PER QUERY)



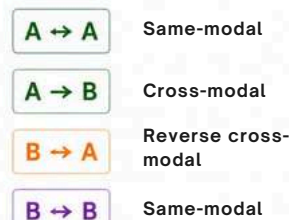
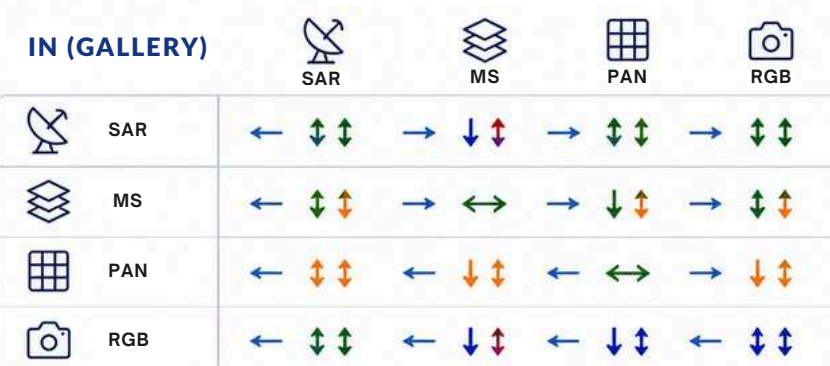
3. EXAMPLE USE CASE

Disaster Response: Flood Mapping



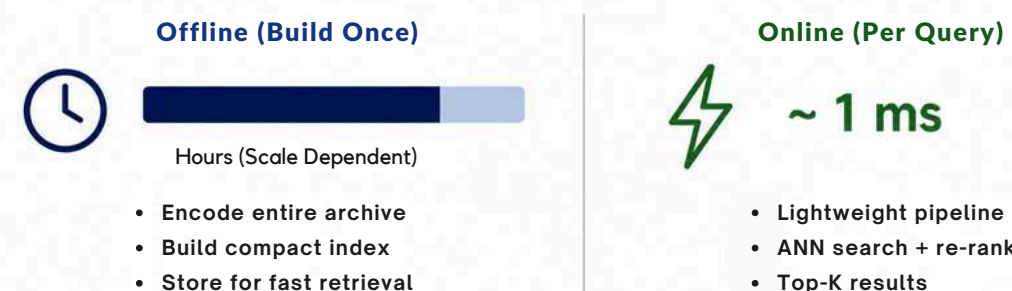
4. FOUR RETRIEVAL DIRECTIONS (A+B)

RETRIEVE FROM (QUERY)



SABER supports all four retrieval directions across all sensors.

5. LATENCY BREAKDOWN (PER QUERY)



Objective: High retrieval accuracy with sub-millisecond latency at million-scale.

6. WORKLOAD & SCALE

Archive Scale



10⁶ - 10⁸ Scenes

Multi-sensor, multi-temporal national archives

Query Load



10³-10⁵ Queries/day

Analysts, systems & automated pipelines

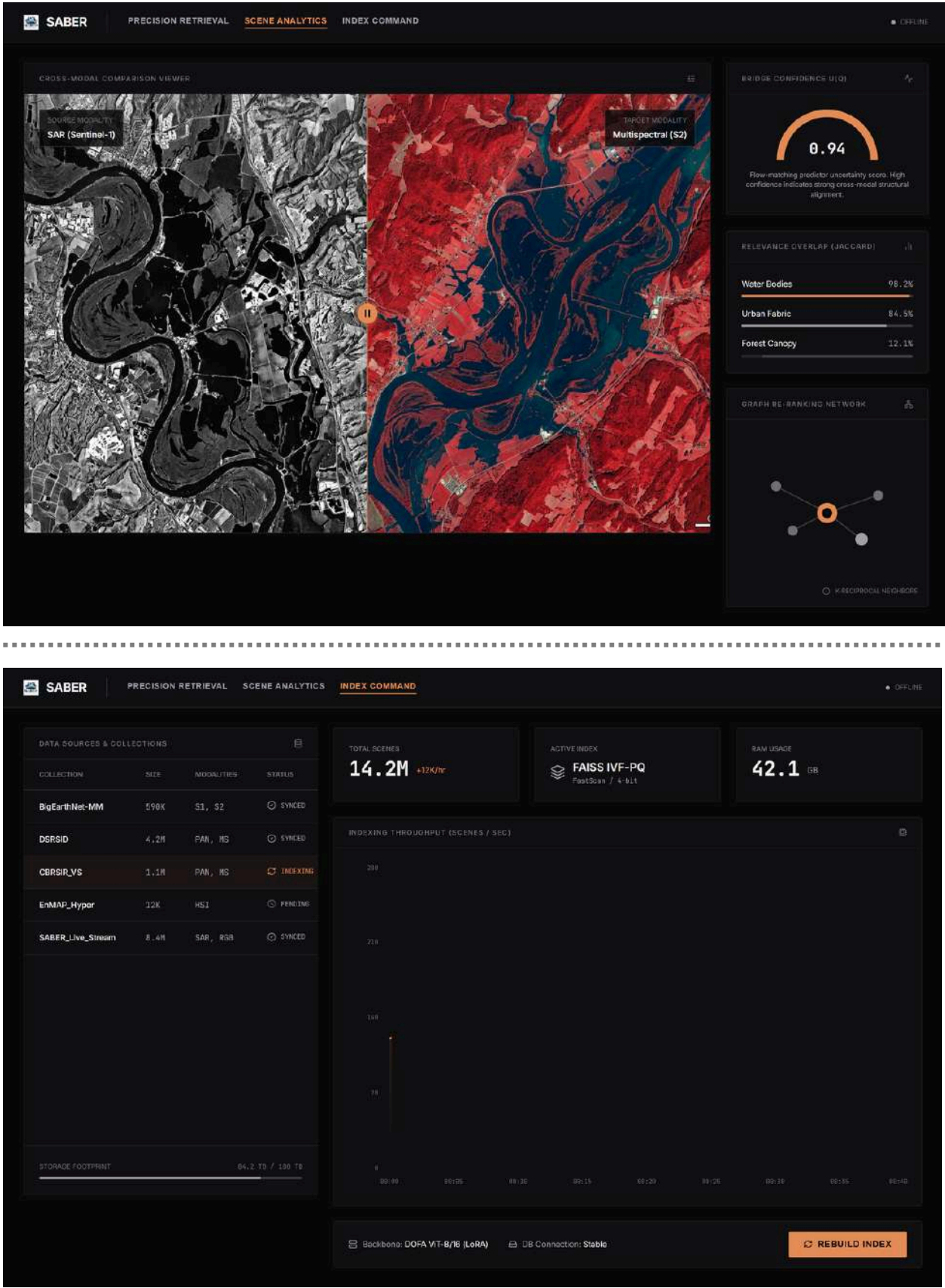
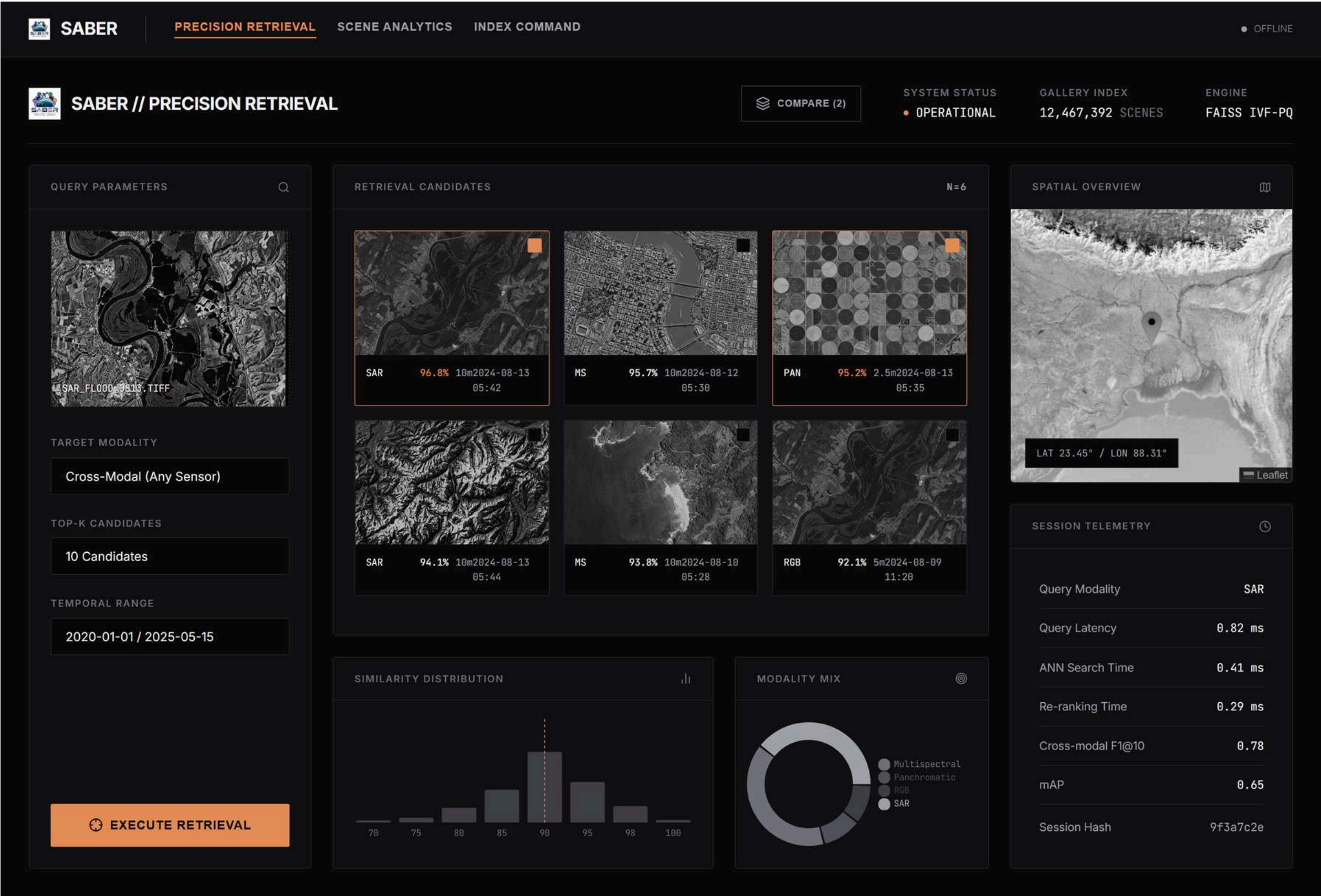


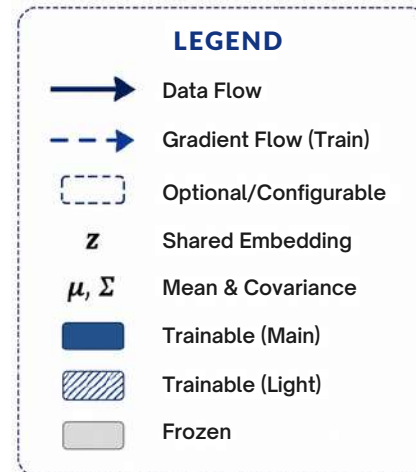
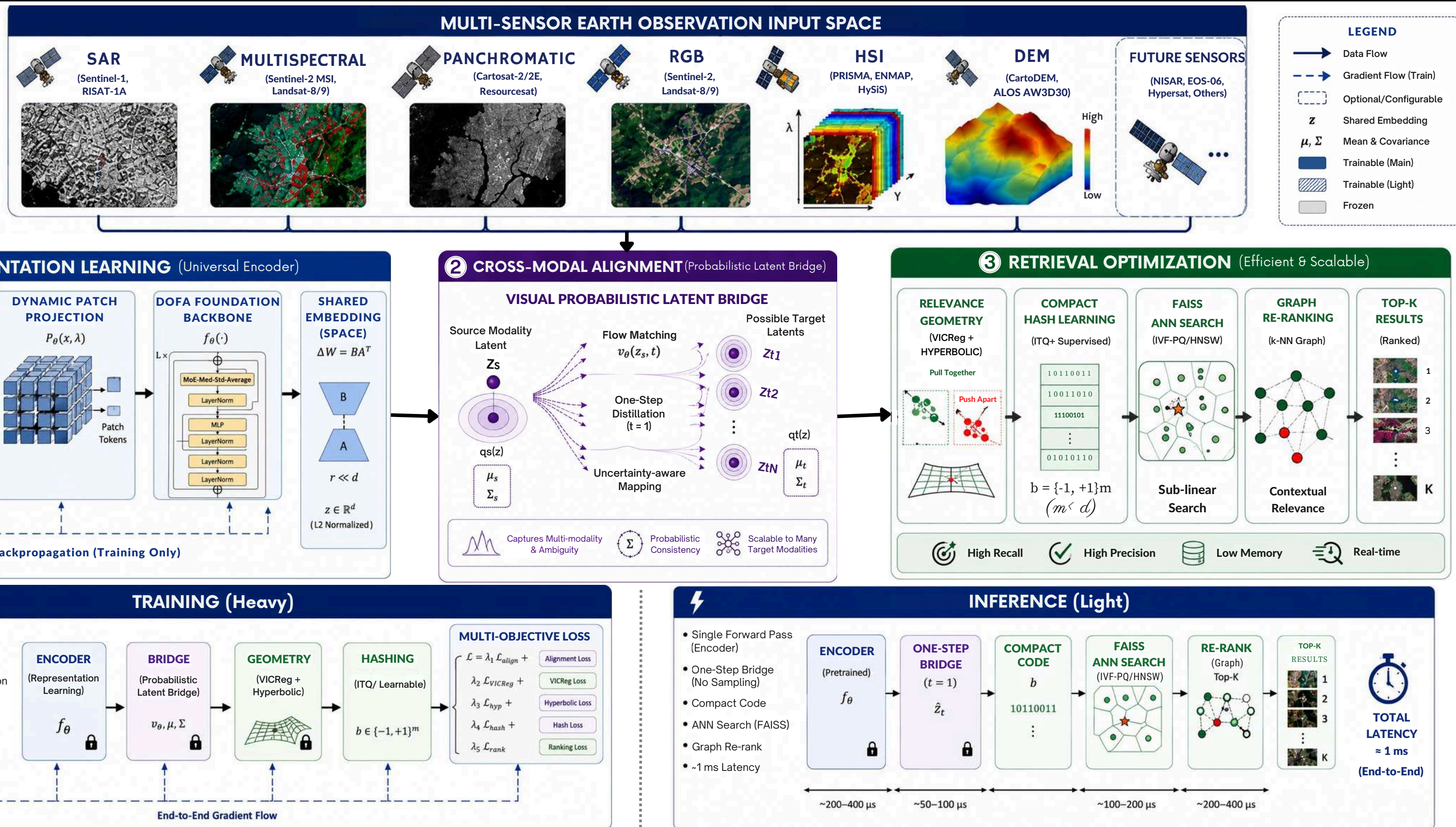
Designed for national-scale EO repositories and real-time operational use.



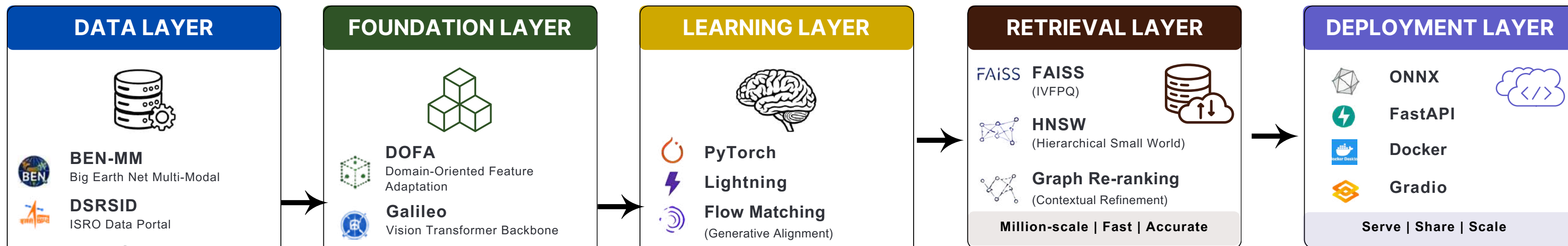
Retrieval Dashboard: Conceptual UI

Illustrative dashboard demonstrating query submission and retrieval of ranked cross-modal results.





SABER TECHNOLOGY STACK (END-TO-END PIPELINE)



PARAMETER EFFICIENCY

Total Parameters : ~1.31B

98%

2%

Frozen Parameters
-1.28B (98%)

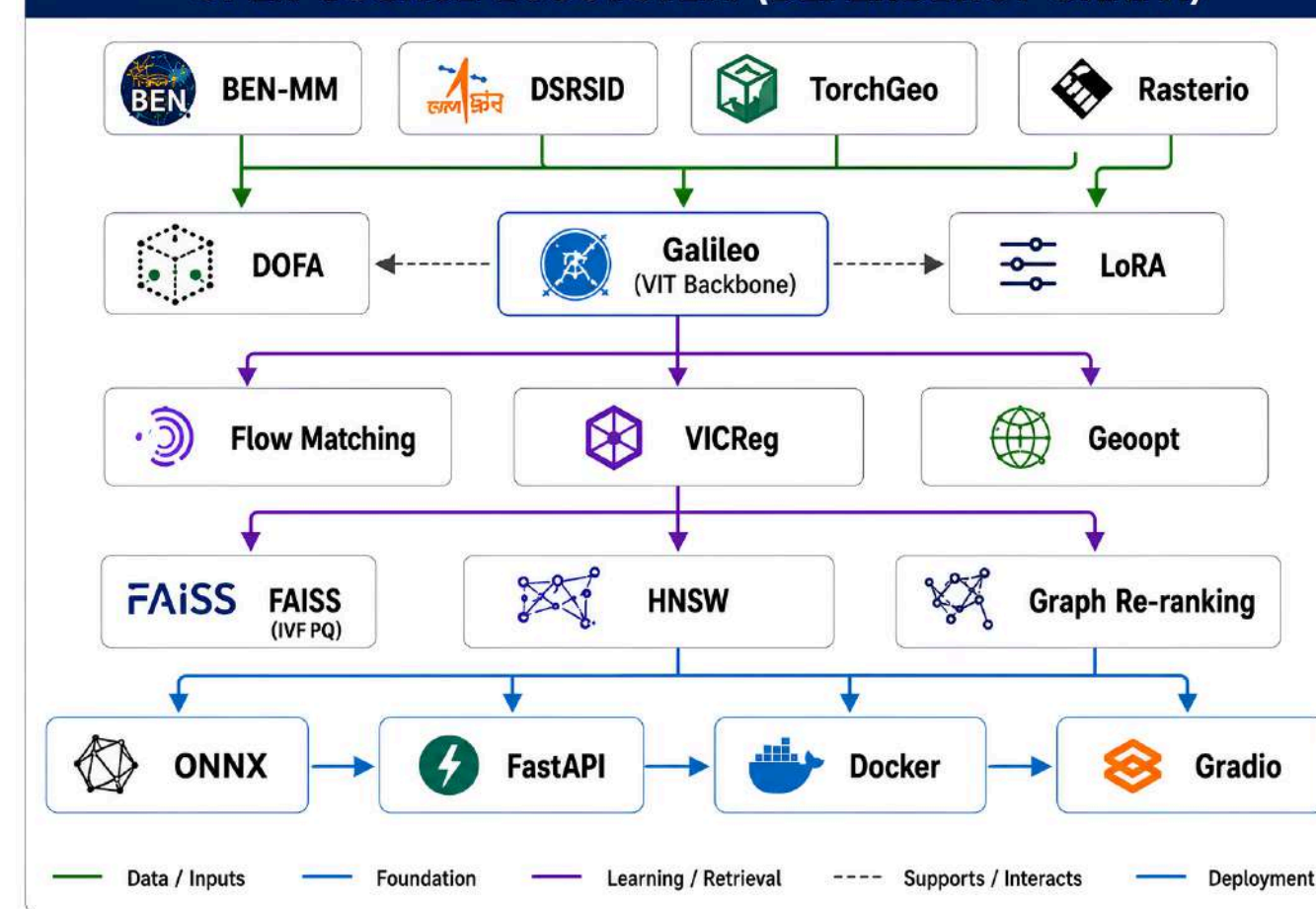
Trainable Parameters (LORA)
-26.2M (2%)

> 49x fewer trainable parameters than full fine-tuning

COMPUTE PROFILE

- 1x NVIDIA A100**
(40 GB)
- Precision**
bf16 (mixed precision)
- Storage**
~120 GB
- Training Time**
1 day (24-30 hrs)
- Trainable Parameters**
<2%
- Frozen Backbone**
98%

OPEN-SOURCE ECOSYSTEM (DEPENDENCY GRAPH)





Project Roadmap and Risk Assessment

Implementation milestones, evaluation strategy, and technical risk management for reliable deployment.



REFERENCES

[1] S. Choudhury et al., "X-JEPA: A Novel Joint Learning Cross-Modal Predictive Alignment Framework for Remote Sensing Image Retrieval," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, 2026. [Link](#)

[2] S. Choudhury et al., "REJEPA: A Joint-Embedding Predictive Architecture for Efficient Remote Sensing Image Retrieval," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, 2025. [Link](#)

[3] Z. Xiong et al., "Neural Plasticity-Inspired Multimodal Foundation Model for Earth Observation (DOFA)," *arXiv preprint arXiv:2403.15356*, 2024. [Link](#)

[4] Y. Lipman et al., "Flow Matching for Generative Modeling," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2023. [Link](#)

[5] A. Bardes, J. Ponce, and Y. LeCun, "VICReg: Variance-Invariance-Covariance Regularization for Self-Supervised Learning," *arXiv preprint arXiv:2105.04906*, 2021. [Link](#)

[6] M. Nickel and D. Kiela, "Poincaré Embeddings for Learning Hierarchical Representations," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2017. [Link](#)

[7] M. Douze et al., "The Faiss Library," *arXiv preprint arXiv:2401.08281*, 2024. [Link](#)

[8] G. Sumbul et al., "BigEarthNet-MM: A Large-Scale Multimodal Multilabel Benchmark Archive for Remote Sensing Image Classification and Retrieval," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 3, pp. 174-180, 2021. [Link](#)

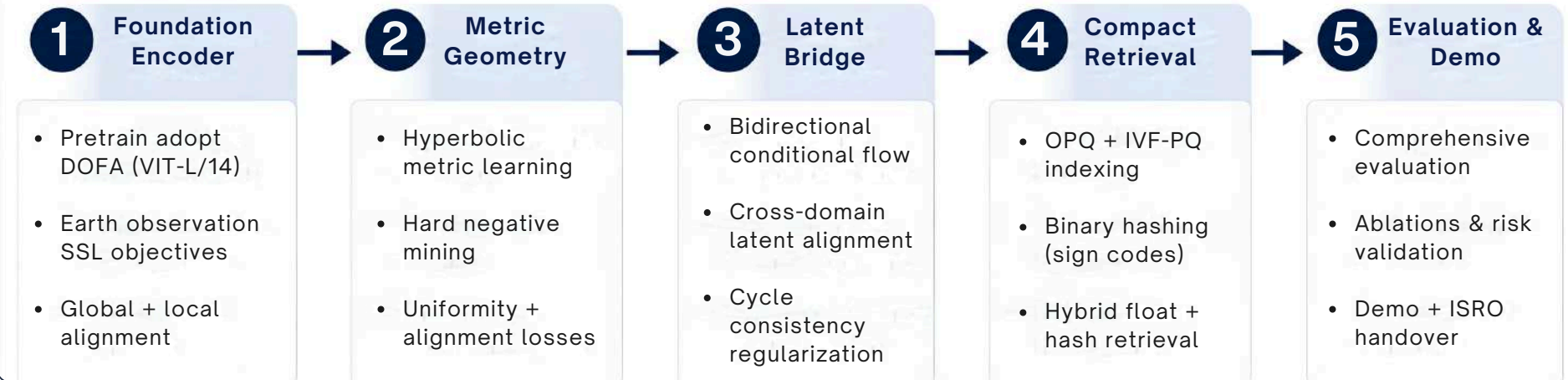


ABLATION STRATEGY

COMPONENT	OPTION A (PROPOSED)	OPTION B (BASELINE)	METRIC IMPACT
Backbone Encoder	DOFA (VIT-L/14)	VIT-L/14 (DINOv2)	Recall@K, nDCG@K
Latent Bridge	Conditional Flow	Deterministic MLP	Recall@K, R@1
Metric Geometry	Hyperbolic	Euclidean	Recall@K, MR-1
Compact Retrieval	Hash (sign) + OPQ	Float IVF-PQ	Recall@K, Latency, RAM
Reranking	Cross-Encoder Rerank	No Rerank (ANN Only)	nDCG@K, R@1, Latency



DEVELOPMENT ROADMAP



SCIENTIFIC RISK MITIGATION

RISK AREA	DESCRIPTION / IMPACT	PRIMARY APPROACH	FALLBACK STRATEGY
Latent Bridge (Flow)	Flow training may be unstable or slow to converge	Conditional normalizing flow	Deterministic MLP mapping
Metric Geometry (Hyperbolic)	Hyperbolic optimization may underperform or be unstable	Hyperbolic embeddings (Poincaré Ball)	Euclidean embeddings (cosine/IP)
Compact Retrieval (Hashing)	Hash codes may lose too much discriminative information	Binary hashing (sign codes) + OPQ	Float FAISS (IVF-PQ) without hashing
Domain Gap	EO SAR/Cross-sensor shift degrades retrieval	Cross-domain alignment + extensive augments	Parameter-efficient LoRA tuning per domain
Reranking	Reranker adds latency or limited gains	Cross-encoder reranker	Pure ANN results(no rerank)



**BHARATIYA
ANTARIKSH
HACKATHON**

2026

THANK YOU